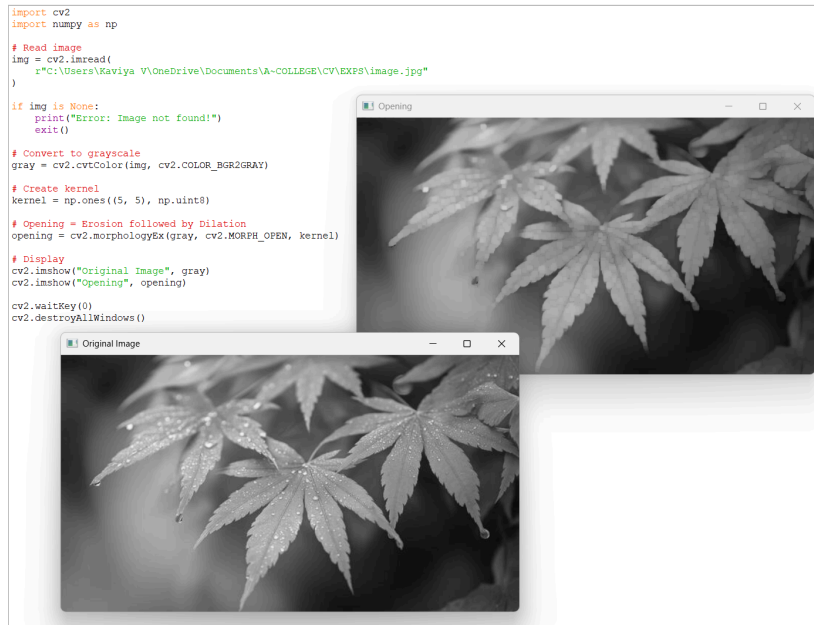


LAB EXPERIMENTS

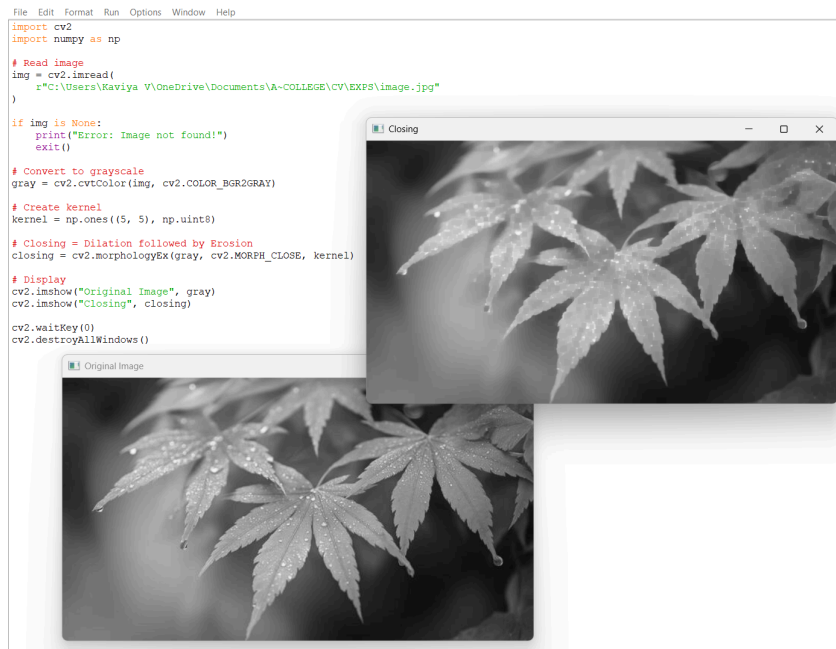
NAME: KAVIYA V
COURSE CODE: ITA0512

REG NO: 192421022
COURSE NAME: COMPUTER VISION

21. Implement the Opening technique as a Morphological operation to dilate the foreground regions based on Open CV.



22. Implement the Closing technique as a Morphological operation to dilate the foreground regions based on Open CV.



23. Implement the Top hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

```
import cv2
import numpy as np

# Read image
img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A-COLLEGE\CV\EXPS\image.jpg"
)

if img is None:
    print("Error: Image not found!")
    exit()

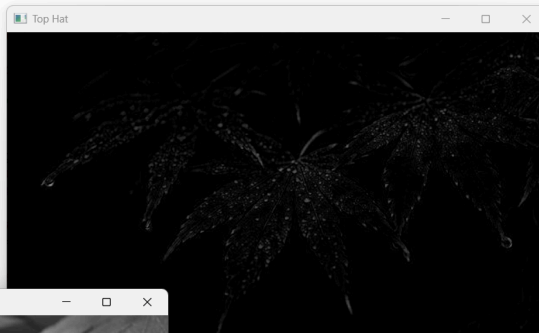
# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Top Hat
tophat = cv2.morphologyEx(gray, cv2.MORPH_TOPHAT, kernel)

# Display
cv2.imshow("Original Image", gray)
cv2.imshow("Top Hat", tophat)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



24. Implement the Black hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

```
File Edit Format Run Options Window Help
import cv2
import numpy as np

# Read image
img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A-COLLEGE\CV\EXPS\image.jpg"
)

if img is None:
    print("Error: Image not found!")
    exit()

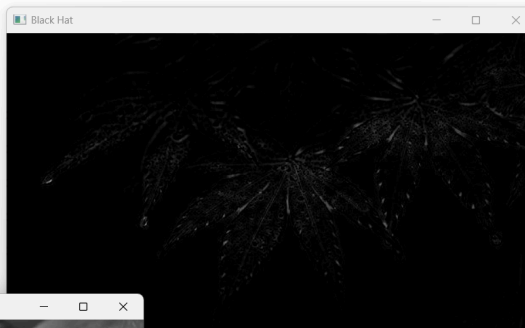
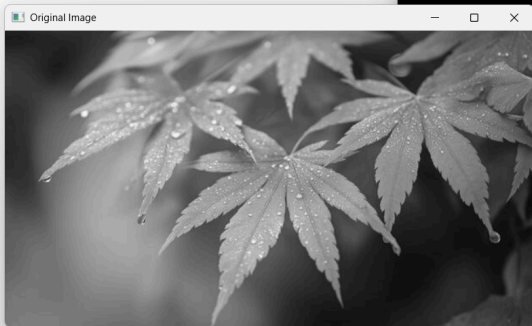
# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Black Hat
blackhat = cv2.morphologyEx(gray, cv2.MORPH_BLACKHAT, kernel)

# Display
cv2.imshow("Original Image", gray)
cv2.imshow("Black Hat", blackhat)

cv2.waitKey(0)
cv2.destroyAllWindows()
```



25. Recognize watch from the given image by general Object recognition using Open CV.

File Edit Format Run Options Window Help

```
import cv2

img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A-COLLEGE\CV\EXPS\image.jpg"
)

if img is None:
    print("Error: image.jpg not found!")
    exit()

# Display image
cv2.imshow("Image", img)

print("Select the watch region using your mouse.")
print("Press ENTER after selecting the region.")
print("Press ESC to cancel.")

# Select ROI
x, y, w, h = cv2.selectROI("Image", img, False)

# Crop watch
watch = img[y:y+h, x:x+w]

# Save cropped watch
cv2.imwrite(
    r"C:\Users\Kaviya V\OneDrive\Documents\A-COLLEGE\CV\EXPS\watch.jpg",
    watch
)

cv2.destroyAllWindows()

print("watch.jpg created successfully!")
```

Image



26. Implement a function to reverse the frames of the video to create a video in reverse mode using Open CV.

File Edit Format Run Options Window Help

```
import cv2

# Read video
cap = cv2.VideoCapture(
    r"C:\Users\Kaviya V\Downloads\video.webm"
)

if not cap.isOpened():
    print("Error: Video not found!")
    exit()

# Get video properties
fps = cap.get(cv2.CAP_PROP_FPS)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))

# Store frames
frames = []

while True:
    ret, frame = cap.read()

    if not ret:
        break

    frames.append(frame)

cap.release()

# Create output video
fourcc = cv2.VideoWriter_fourcc(*"mp4v")

out = cv2.VideoWriter(
    "reverse_video.mp4",
    fourcc,
    fps,
    (width, height)
)

# Write frames in reverse order
for frame in reversed(frames):
    out.write(frame)

out.release()

print("Reverse video created successfully.")
```

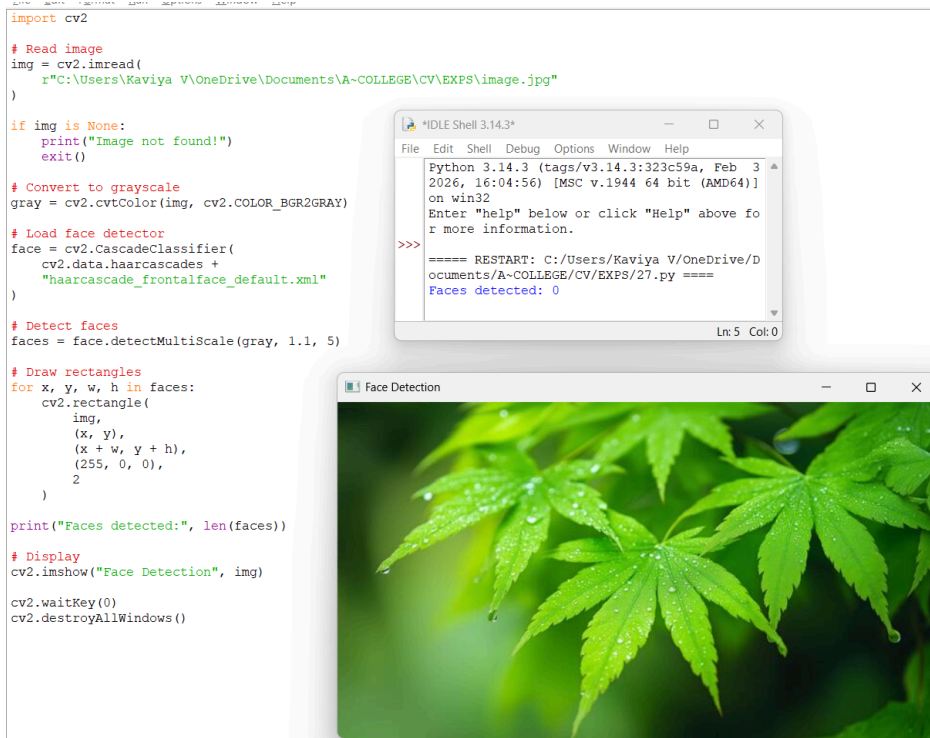
IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

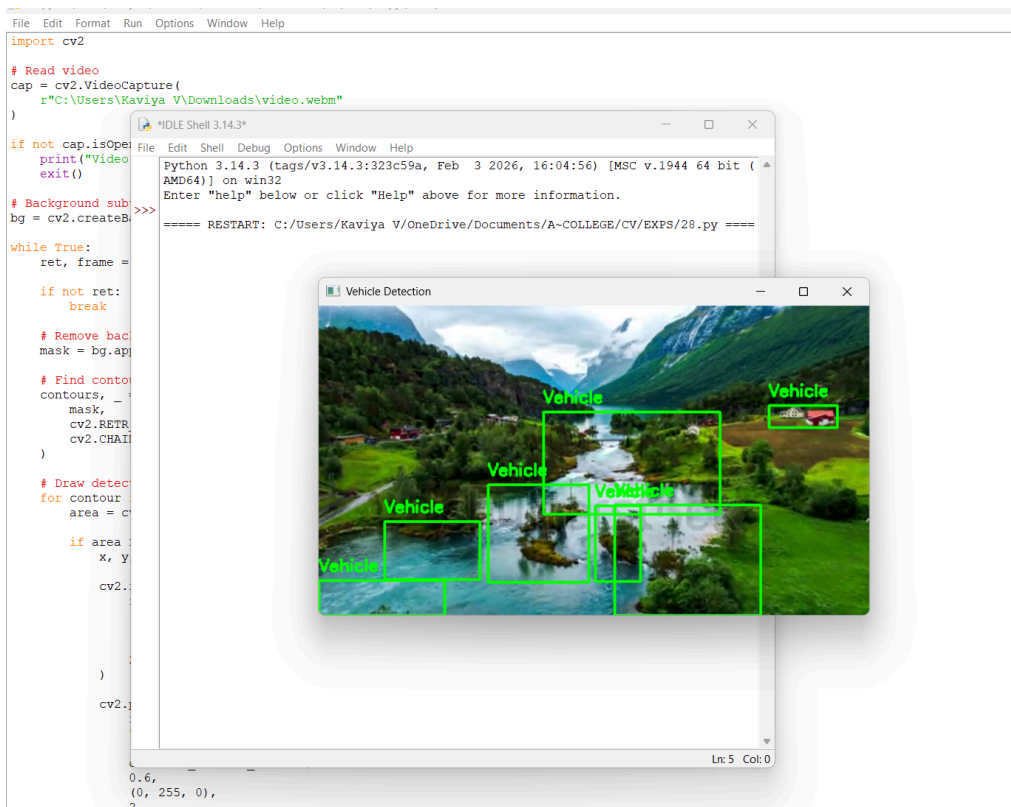
```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/Kaviya V/OneDrive/Documents/A-COLLEGE/CV/EXPS/26.py =====
>>> Reverse video created successfully.
```

Ln: 6 Col: 0

27. Implement a face detection algorithm using Open CV to detect and locate human faces in the images.



28. Implement a vehicle detection algorithm using Open CV to detect and locate vehicles in each frame of the video.



29. Implement an Eye detection algorithm using Open CV to detect and locate human eyes in the images.

```
File Edit Format Run Options Window Help
import cv2

# Read image
img = cv2.imread(
    r"C:\Users\Kaviya V\OneDrive\Documents\A-COLLEGE\CV\EXPS\image.jpg"
)

if img is None:
    print("Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Load eye detection cascade
eye = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_eye.xml')

# Detect eyes
eyes = eye.detectMultiScale(gray,
                             1.1,
                             5)

# Draw rectangles around eyes
for (x, y, w, h) in eyes:
    cv2.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), 2)

print("Eyes detected: ", len(eyes))
cv2.imshow("Eye Detection", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1914 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

==== RESTART: C:/Users/Kaviya V/OneDrive/Documents/A-COLLEGE/CV/EXPS/28.py ====
==== RESTART: C:/Users/Kaviya V/OneDrive/Documents/A-COLLEGE/CV/EXPS/29.py ====
Eyes detected: 0
==== RESTART: C:/Users/Kaviya V/OneDrive/Documents/A-COLLEGE/CV/EXPS/29.py ====
Eyes detected: 0



Ln: 10 Col: 0

30. Implement a Smile detection algorithm using Open CV to detect and locate human smile in the images.

```
File Edit Format Run Options Window Help
print("Image not found!")
exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Load face and smile cascade classifiers
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')
smile_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_smile.xml')

# Detect faces
faces = face_cascade.detectMultiScale(gray,
                                       1.1,
                                       5)

# Detect smiles
for (x, y, w, h) in faces:
    face_gray = gray[y:y+h, x:x+w]
    smiles = smile_cascade.detectMultiScale(face_gray,
                                              1.1,
                                              20)

    for (sx, sy, sw, sh) in smiles:
        cv2.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), 2)

print("Faces detected: ", len(faces))
cv2.imshow("Smile Detection", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1914 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

==== RESTART: C:/Users/Kaviya V/OneDrive/Documents/A-COLLEGE/CV/EXPS/30.py ====
Faces detected: 0



Ln: 5 Col: 0